

Testing Significance of Moderator Effects on Multiple Subject Behavior Networks

In psychological research, there is a pressing interest in understanding how socioeconomic status, physical health status, and prior mental health diagnoses may influence the emotional and behavioral patterns of individuals. A deeper understanding of these influences can aid in the development of comprehensive psychological profiles for subject assessment and can ultimately inform personalized psychological interventions [1]. Motivated by these analytical needs, the goal of this project is to develop and make available a toolkit of novel statistical models and efficient algorithms to quantify and test the impact of between-person differences in the behavioral patterns of individuals over time.

This project builds on an ongoing collaboration between the Departments of Psychology and Statistics at UCR [2], where I aim to use multi-subject longitudinal data to estimate Granger-causal networks that describe how depression, anxiety, stress, quality of sleep, and physical activity influence each other. An example of multi-subject Granger-causal networks is found in Figure 1. Nodes in the networks correspond to behavioral constructs, and links describe Granger-causal effects, *i.e.*, whether past values of one construct predict future values of another [3]. For example, a directed link from sleep quality to anxiety suggests that sleep quality predicts changes in next-day anxiety levels. Each of these links can potentially be affected by *moderators*, which are additional constructs that influence the strength of the Granger-causal effects.

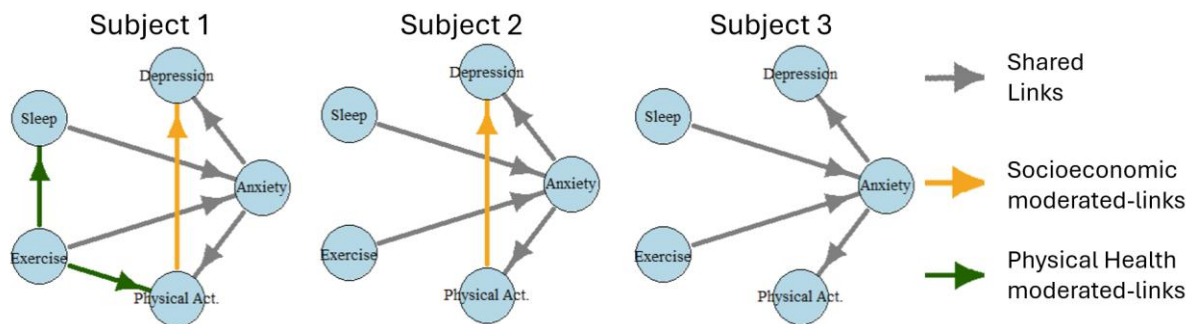


Figure 1: Example of Granger-causal networks across 3 subjects. Each subject has a different behavioral network. Our aim is to quantify whether different network links are moderated by external socioeconomic status or health status.

Prior methods in the literature can perform the estimation of such Granger-causal networks on multi-subject longitudinal data [3-6], and test network significance for a single subject [7-9]. However, these methods either do not incorporate moderator information [3-9], leading to false discoveries, or cannot test whether the estimated networks and moderator effects are statistically significant [3-6], leading to potentially biased results. The development of these tests is challenging because longitudinal observations are highly correlated, reducing the effective sample size [3]. Additionally, analyzing heterogeneous subjects increases model complexity, with the number of parameters growing combinatorially with the number of variables. For example, for 10 moderators, 10 variables in the Granger-causal network, and 50 subjects, we must assess a total of 22,500 parameters. Performing reliable inference in these high-dimensional settings with correlated observations is a substantial statistical and computational challenge.

In my collaborative project at UCR, I have developed the moderator vector autoregressive model (MOD-VAR), a new method that estimates individual Granger-causal networks and quantifies the effect of moderator variables on the network links of each subject [2]. MOD-VAR is effective in high-dimensional settings, as confirmed via theoretical guarantees and extensive numerical experiments. This work is currently being prepared for submission to *Psychological*

Methods. Furthermore, I have extensive experience with the estimation of structures across multiple network models [2,10-12].

Objectives

The core objective of this proposal is to develop a statistically rigorous hypothesis testing framework based on the MOD-VAR to assess, for each potential network edge and moderator variable, whether the moderator significantly influences Granger-causal effects across subjects. The effectiveness of the testing procedure will be verified through theoretical analysis, numerical experiments, and real-data applications. I aim to develop a software implementation in R, making this toolkit widely accessible for analyzing multi-subject longitudinal data. This work will be developed in collaboration with collaborators from the Department of Psychology and a graduate student from the Department of Statistics at UCR.

I aim to expand MOD-VAR beyond behavioral longitudinal studies through collaborations with interdisciplinary research teams at UCR and other institutions. I am establishing a collaboration with researchers from the Department of Psychology at UCR, and the Department of Mathematics at the Technical University of Munich, to develop and apply similar testing procedures to measure the moderator effect that brain region sizes have on the Granger-causal structure of brain networks across multiple subjects.

Finally, I plan to apply for extramural funding to support the research activities of these extensions.

Specific aims

[A1] Develop and implement the significance-testing procedure [Months 1-6]: Develop the statistical software for multiple testing across moderator-network edges, accounting for high dimensionality and subject heterogeneity. I aim to implement the testing method and release it as an R package.

[A2] Evaluate performance [Months 7-12]: Assess the testing procedure on simulated and real data, with comparisons to existing methods in the literature. Apply the procedure to the GLOBEM dataset [13], as well as behavioral data owned by collaborators, to examine the influence of socio-economic moderators on behavioral patterns over time.

[A3] Establish interdisciplinary extensions [Months 13-24]: Disseminate results at national and international conferences. Furthermore, establish new interdisciplinary collaborations and prepare proposals for extramural funding.

Anticipated Results

I anticipate the development of the manuscript "*Testing Moderator Effects on Multiple-Subject Granger-Causal Networks on Intensive Longitudinal Data*," together with a companion R package. In addition, I expect the establishment of new collaborative teams to extend MOD-VAR to general multi-subject longitudinal studies. I aim to extend MOD-VAR for the analysis of brain connectivity networks using fMRI data to determine how moderators influence brain connectivity pathways [14,15]. This project would be in collaboration with faculty at the Department of Psychology at UCR, and Prof. Marie Düker at the Technical University of Munich. These extensions will position the project for competitive extramural funding. The Regents Faculty Fellowship will provide early support for the development of the MOD-VAR methodology and generate preliminary results on neuroimaging data, which are vital for future grant applications and the long-term advancement of my research program.

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